JEDDAH TOWER (SAUDI ARABIA) SAVC Entry Dossier

Media Façade / Architectural LED + Control

Prepared for:
SAVC (Star Asia Vision Corporation)

Research entry date (UTC):
11 Jan 2026

Research end date (UTC):
24 Jan 2026

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Research Focus:

SAVC Media Façade / Architectural LED + Control/CMS entry via local KSA partnership and a façade-integration route through **JANGHO** (curtain wall package), with design-team acceptance through **AS + GG + FMS**, and formal submittals through the delivery chain **(TURNER / SBG / DAR)**.

All external sources were accessed on the research entry date unless otherwise stated.

1. Executive Snapshot (Decision-Ready):

Key Takeaway:
Project restart and delivery-chain activity are confirmed: JEC signed a completion agreement with SBG (SAR 7.2bn; 42-month period), Jangho commenced curtain wall works with the first unit installed on 30 Nov 2025, and KONE publicly identifies Turner as project management and DAR as engineering support. SAVC's highest-probability entry is an embedded media facade system routed via Jangho, with parallel design-team acceptance via AS+GG and Fisher Marantz Stone.

Opportunity Posture:
Landmark national asset; facade package is active (Jangho), creating a practical gateway for embedded technology.

Best Route:
Dual-track approvals (Jangho technical + AS+GG/FMS design acceptance) while aligning submittals with Turner/DAR governance.

Immediate Gates:

1. Sponsor access
2. Confirmation that facade lighting/media scope is still buyable (new package or VO)
3. Interface allowances in curtain wall shop drawings.

2. Verified roles (What Each Party is on the Project)

1. JEC (Jeddah Economic Company): Developer/Owner.
2. SBG (Saudi Binladin Group): Principal Contractor.
3. Turner (Turner Project Management / TiME / Turner Arabia): Project Management.
4. DAR = Dar Al-Handasah (Shair & Partners): Engineering Support.
5. AS+GG (Adrian Smith + Gordon Gill Architecture): Architect.
6. Jangho / Jangho Arabia Construction L.L.C.: Façade Contractor (Unitised curtain wall).
7. FMS (Fisher Marantz Stone): Lighting Consultant is widely cited in project references; treat as "influencer gate" until reconfirmed in the current design team directory.

3. Decision Power Map (Who Decides vs Who Influences): SAVC Lens

A. Decides (Commercial / Package Award / Access)

- **JEC:** ultimate sponsor for add-scope / VO / brand-led façade media intent (budget, owner sign-off).
- **SBG:** executes procurement on site; can block/unblock access and nominated-sub pathways.
- **Jangho (within façade package):** decides practicality of embedding SAVC tech into curtain-wall units (panel method, shop drawings, QA/QC, maintainability).

B. Influences (Technical Acceptance / Spec Compliance / Routing)

- **Turner:** controls submittal routing, governance, schedule gates, and "what gets reviewed when."
- **Dar Al-Handasah:** engineering acceptance and compliance alignment for façade-integrated systems (interfaces, loads, safety, maintainability).
- **AS+GG + FMS:** design intent + performance criteria; they can enable "Approved Alternate / BoD acceptance" and set mock-up requirements.

4. Verified Project Facts (Evidence Table)

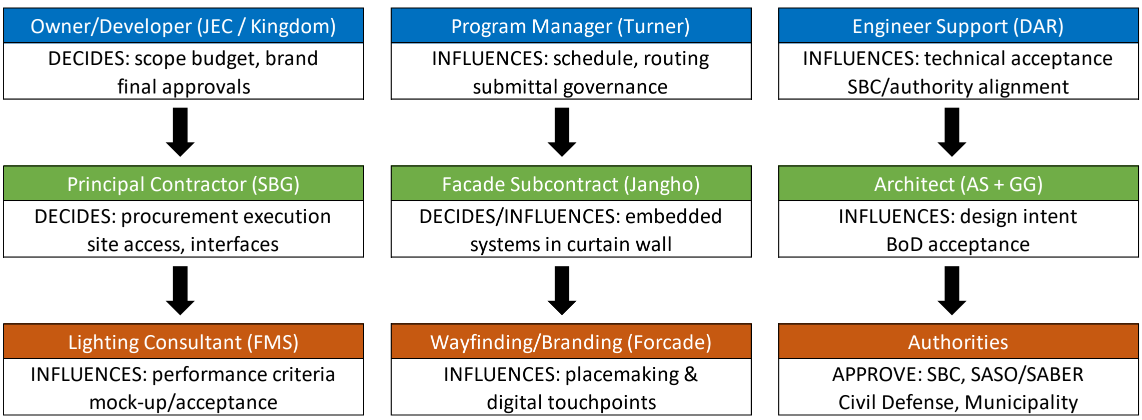
Key Takeaway:
Restart contract, named delivery parties, and facade commencement are confirmed by primary sources. Unknowns are explicitly flagged as [To verify].

FACTS	SOURCE URL	PUBLISHED	CONFIDENCE
JEC signed agreement with SBG to resume construction; contract value SAR 7.2bn; anticipated period 42 months.	https://kingdom.com.sa/news/jeddah-economic-company-has-signed-an-agreement	02-Oct-24	High
Jangho won curtain wall sub-contract (bid amount SAR 1.052bn) and provided technical services after restart.	https://en.jangho.com/JANGHONEWS/2025/1023/134.html	23-Oct-25	High
Jangho installed the first curtain wall unit on 30 Nov 2025; facade construction officially commenced.	https://en.jangho.com/JANGHONEWS/2025/1203/140.html	03-Dec-25	High
KONE: AS+GG architect; Turner project management; DAR engineering support (public release).	https://www.kone.com/en/news-and-insights/releases/kone-to-equip-jeddah-tower--the-world-s-tallest-building-to-be-in-saudi-arabia-with-people-flow--solutions-2025-10-08.aspx	08-Oct-25	High
CTBUH listing includes Fisher Marantz Stone (lighting) and Forcade (wayfinding).	https://www.skyscrapercenter.com/jeddah/kingdom-tower/2	Listing (Accessed)	Medium
Forcade project page: 'Jeddah Tower Development Wayfinding' (exterior, base building, digital).	https://forcadedesign.com/project/jeddahtower	Undated (Active)	Medium

5. Stakeholder 'Who-is-Who' and Decision Power Map

Key Takeaway:
Final scope/budget approvals sit with the owner/developer; procurement execution sits with SBG; embed-able facade technology is effectively gated by Jangho. Design and performance acceptance runs through AS+GG and FMS, with

Decision Power Map:



6. What Scopes Are Still Buyable and Where SAVC Fits

Key Takeaway:
Curtain wall is awarded and underway. SAVC must target scopes that can be embedded/nominated under Jangho's unitised curtain wall system and/or issued as dedicated facade-lighting/media-facade packages.

SCOPE	ASSESSMENT	ROUTING PARTY	FIT / STATUS
Facade-integrated architectural LED (linear + pixel nodes)	Best buyability if mounting/cable pathways are still open in curtain wall shop drawings; otherwise becomes retrofit with high risk/cost.	Primary route: Jangho -> SBG (nominated subsystem)	High
Defined media facade zones (podium / crown / spire accents)	Often VO-able once branding intent is confirmed; may be packaged with control/CMS late in schedule.	Primary route: JEC/Turner (scope) + Jangho (execution).	Medium - High
Control/CMS + monitoring + governance	Often procured later; can be separated if interfaces are defined (DMX/Art-Net edge, IP backbone).	Primary route: Turner/DAR + local integrator.	Medium
Prior media-façade proposals/submittals	Unknown. Request any historic submittals, alternates, or concept packages already reviewed.	Request from Turner (data room) + FMS + SBG.	[To Verify]

Detailed scope positioning (SAVC)
SAVC should pitch “Media Façade-ready Architectural Lighting” as a façade system enhancement, not as a standalone “screen”:

- Primary scope:
  Linear + node-based façade luminaires integrated into unitized panels; corrosion-resistant, serviceable from interior; designed for 42-month schedule constraints; factory QA/QC aligned with JANGHO panel production.

- Control:
  Centralized control servers + distributed nodes; show mode operation, schedule, event triggers, remote diagnostics, redundancy.

- Optional media layer:
  Select façade zones that can run low-resolution pixel content for national days/events/branding moments.

7. SAVC-Specific Scope Map (Work Packages, Interfaces, Risks, Value-Adds)

Key Takeaway:
To be accepted, SAVC must bring interface clarity: where SAVC ends, where JANGHO begins, and how approvals flow through FMS/AS+GG → DAR → TURNER → SBG.

A. Work packages (Proposed)

WP-1: Media Façade Lighting Hardware
Linear/tube luminaires, pixel nodes, mounting channels, brackets, connectors.

WP-2: Media Façade Control System
Drivers, decoders, data distribution, network switches, gateways.

WP-3: Central CMS / Show Control
Scheduling, scenes, pixel mapping, monitoring dashboards, user roles.

WP-4: Engineering & Submittals
Shop drawings support, method statements, ITP/QCP, calculations, testing plans.

WP-5: Mockups & Testing
Sample panel mockup at JANGHO factory + on-site night test.

WP-6: Commissioning + Training
Handover, O&M manuals, staff training, as-builts.

WP-7: O&M + SLA (Optional but strategic)
24/7 monitoring, quarterly inspections, spares management.

B. Interfaces (who owns what)

INTERFACE	OWNER	SAVC ROLE	RISK IF UNCLEAR	SAVC VALUE-ADD
LED integration in unitized panels	JANGHO	Provide integration details, tolerances, service access logic	Panel redesign late; water ingress	"Panel-friendly" design; minimal penetrations; defined service hatch logic
Design intent / aesthetics	FMS + AS+GG	Provide photometrics, glare control, viewing studies	Rejection in mockup	Full mockup plan + alternate optics + dimming curves
Local compliance & approvals	DAR	Provide data sheets, certifications, test reports	Submittal delays	Compliance pack aligned to SBC + SASO/SABER
Procurement & commercial route	SBG	Provide pricing, lead times, warranty, spares	Delayed award	Package options + schedule-aligned delivery plan
Governance / documentation	TURNER	Conform to submittal workflow, milestones	Gate reviews fail	"Turner-ready" submittal matrix + RFI log discipline

C. Where Risks Concentrate

Schedule lock-in:
Unitized panels require early freeze; late lighting changes become high-cost.

Environmental durability:
Red Sea coastal exposure (salt + humidity + heat).

Access/maintenance:
Megatall façade maintenance must be designed-in (service corridors, replaceable segments, standardized modules).

Approvals:
Mockup acceptance is the decisive event.

8. Procurement Status (Confirmed vs Must-Request) + Access Routes

Key Takeaway:
Public sources confirm the macro contract and the delivery chain; package-level procurement visibility is not public and must be obtained via a targeted data-room request through SBG/TURNER.

Confirmed (publicly)

JEC↔SBG agreement: SAR 7.2B (~AED 7.05B | USD 1.92B), 42 months.
Delivery chain including TURNER & DAR confirmed by KONE.
Curtain wall scope is actively held by JANGHO (design/technical service and unitized curtain wall area disclosed by JANGHO)

Must-request (from whom)

WHAT WE NEED	WHY IT MATTERS	WHO TO REQUEST FROM	ASK FORMAT
Package procurement plan (façade lighting, show control, façade access systems, ELV)	Identify what's still buyable	TURNER (controls the master procurement plan)	"Package status matrix request"
JANGHO panel shop drawing schedule + façade mockup plan	Determine integration window	JANGHO (façade delivery owner)	Technical workshop + NDA
Lighting basis-of-design + control narrative	Determine acceptance pathway	FMS (lighting) + AS+GG	Alternate/BOD proposal + mockup plan
Submittal routing protocol & document templates	Avoid rework	TURNER	Request latest submittal register + workflow
Vendor registration + prequal requirements	Entry permission	SBG	Vendor onboarding package

9. Practical access routes (highest probability)

Design-led entry (recommended):

AS+GG + FMS acceptance → then route to DAR/TURNER for inclusion → commercialize via SBG.

Façade-led entry:

JANGHO partnership (as their nominated LED/control subsystem) → then seek FMS/AS+GG acceptance via façade mockup.

Contractor-led entry:

SBG vendor registration → bid package if opened → harder without design pull.

10. Secure Design-Team Acceptance (AS+GG + FMS) and Route Submittals

Key Takeaway:
Secure BoD or approved alternate acceptance with AS+GG and FMS, then route formal submittals through DAR/Turner to SBG and Jangho.

- Weeks 1 to 3:
  Performance-based concept pack to AS+GG + FMS (photometrics, IP/thermal, corrosion, control topology).

- Weeks 4 to 8:
  Integration workshop with Jangho (mounting, wiring, access, responsibilities).

- Weeks 9 to 15:
  Submit Approved Alternate pack via DAR/Turner templates; include compliance plan (SABER) and mock-up plan.

- Mock-up:
  Unitised panel mock-up with embedded SAVC components for acceptance tests.

Submittal Routing:
AS+GG/FMS endorsement -> DAR acceptance -> Turner governance -> SBG procurement -> Jangho integration + install -> SAVC commissioning -> SAT + handover

11. Regulatory and Compliance Checklist (KSA)

Key Takeaway:
SABER/SASO compliance for imported electrical/lighting products is mandatory; SBC alignment must be documented for embedded installations; Civil Defence and municipal permits must be coordinated early.

AREA	CHECKLIST ITEMS	REFRENCES
Saudi Building Code (SBC)	SBC alignment for electrical/fire/structural interfaces; document penetrations and firestopping; DAR alignment.	https://sbc.gov.sa/
SASO / SABER compliance	Register products on SABER; plan PCoC/SCoC and test reports; HS-code mapping	https://saber.sa/ and https://www.saso.gov.sa/en/Pages/default.aspx
Saudi Civil Defense	Fire safety approvals and material compliance; coordinate facade-mounted enclosures.	https://998.gov.sa/
Municipality / Balady	External installation permits; signage/advertising/show rules; confirm with local municipality.	https://balady.gov.sa/

Detailed compliance pack (what SAVC should prepare before first formal meeting)

- **Product compliance (SASO/SABER)**
  Product category mapping (LED luminaires, drivers, control gear, power supplies, network equipment).
  Test reports: EMC, electrical safety, IP rating, thermal, corrosion resistance, aging/burn-in.
  Country of origin documents, HS codes, labeling/Arabic requirements.

  (Official SASO/SABER portals should be used for actual registration workflows; align with your import partner and local compliance agent.)

- **SBC Alignment Memo**
  External lighting safety: glare, strobing limitations (particularly if "media" content is planned).
  Cable routing, penetration sealing, and façade fire-stopping coordination with façade engineer and fire consultant.
  Structural attachment notes (loads, vibration, wind).

- **Civil Defence Coordination**
  Define emergency override modes (all-on, all-off, safe scenes).
  Cable fire rating requirements and routing constraints.
  Ensure no conflict with façade smoke management or fire shutters (if any).

- **Municipality / Public Realm**
  If any part becomes advertising DOOH, treat as a separate workstream with licensing and content governance [To verify exact authority path for Jeddah tower façade content].

12. Partner Map - KSA Shortlist (Integrators / Installers / Support)

Key takeaway:
For credibility in KSA, pair SAVC with (a) a local systems integrator capable of ELV/network + commissioning and (b) a façade execution route (JANGHO) for panel integration. Fractal Systems can be viable if their KSA references match façade-scale deployments and they can mobilize to Jeddah.

A. Partner roles needed (minimum viable consortium)

Façade package owner: JANGHO (already on the project).
Local AOR/compliance interface: DAR (already on project).
Local integrator (ELV/control/network): Candidate: Fractal Systems (subject to reference verification).

B. Fractal Systems — can we partner?

Fit assessment (what must be true to proceed):

- They must demonstrate KSA-based commissioning capability, control system integration, and high-rise façade/landmark references with documented scope.
- They must accept SAVC control architecture, documentation discipline, and site staffing requirements.

Recommendation: Proceed to an NDA + capability meeting if they can supply:

- 3 relevant KSA references (project name, client, scope, value band, contacts)
- ELV/network team org chart
- Maintenance/SLA model for mission-critical façade systems

[To verify]
Need direct documentation from Fractal and/or public reference pages: Precise corporate identity, ownership, and KSA project list.

C. JANGHO partnership — Highest Leverage

JANGHO's own disclosures strongly suggest they control the curtain wall engineering narrative and have already executed performance testing for the system.

Therefore, JANGHO is the logical "carrier" for SAVC tech if SAVC positions as:

- Panel-integrated, factory-testable modules
- Serviceable from interior
- Minimal disruption to JANGHO fabrication

13. Draft Technical Concept (SAVC) Performance, Control, Maintenance, Phasing

Key Takeaway:
The technical concept should be engineered as a façade subsystem: corrosion-resistant, IP-rated, thermally derated for Jeddah, and designed for unitized panel production and replacement.

A. Environmental / Durability Assumptions (Jeddah Coastal)

Heat: Design for sustained high ambient; derate drivers; specify thermal protection.
Humidity + Salt: Marine-grade fasteners, coated PCBs, sealed connectors; corrosion testing.
Dust/Sand: Sealed optics and gaskets; cleaning plan.

B. Reference Performance Ranges (To Frame Discussions)

These are reference ranges for concepting and budgeting; final values must be confirmed against FMS design intent, viewing distances, and façade mockup outcomes.

- **Media Façade / Pixel Capability (select zones)**

  Pixel pitch (concept):
  Far-view façade identity: ~50–120 mm pitch
  Mid-view zones (podium/plaza): ~25–60 mm pitch

  Luminance (Night/Show):
  Typically controlled to minimize glare; design for dimmable operation with scene-based caps.

- **Architectural linear / node lighting**

  High uniformity, smooth gradients;
  Glare-controlled optics;
  Tight binning for color consistency.

  Ingress Protection:
  Target IP66–IP67 for exposed façade components; define compartmentalization in unitized panels.

C. Control topology (DMX/Art-Net/eCue-style architecture)
(The Architecture to be proposed by SAVC)

D. Maintenance strategy (must be designed in)
(To be proposed by SAVC)

E. Phasing Plan (Aligned to Unitized Façade Reality)
(To be proposed by SAVC)

14. Competitive Context

Key Takeaway:
Treat SACO as a direct competitor for GCC media façade / architectural LED packages; the winning differentiation is façade-integration readiness (JANGHO route), design-team acceptance, and lifecycle operability (monitoring + SLA).

Competitor Set (Project - Relevant):

- SACO:
  Competitor for media façade / architectural LED scope.
- Other likely competitors [To verify]:
  Global façade lighting brands, large AV integrators, and lighting OEMs typically specified by design consultants.

Differentiators SAVC should emphasize (in this project):

- JANGHO-friendly integration (unitized panel design constraints understood)
- Mockup-led acceptance (de-risking for FMS/AS+GG)
- Red Sea durability spec (coastal corrosion strategy)
- Operational excellence (monitoring dashboards, spares strategy, repair time metrics)

15. Risk Register (With Mitigations)

Key Takeaway:
The two existential risks are (1) being too late for façade integration, and (2) failing mock-up acceptance with FMS/AS+GG.

Risk	Impact	Likelihood	Mitigation
Late entry vs curtain wall design freeze	Loss of integration role	Medium to High	Engage JANGHO immediately; propose "minimal-change integration kits"; lock mockup date fast
FMS/AS+GG rejection at mockup	Scope lost	Medium	Deliver a strong BOD/Alternate pack; offer multiple optics + dimming curves; run pre-mockup lab tests
Procurement opacity under SBG	Delays	High	Vendor registration + Turner procurement-plan request + submittal-ready documentation
Coastal corrosion failures	Warranty risk	Medium	Conformal coating, marine hardware, sealed connectors, accelerated corrosion tests; define cleaning cycles
Maintenance access impractical	Long-term failure	Medium	Design for interior service; modular replacements; spares policy; integrate access strategy early
"Media façade" political/branding sensitivity	Scope reduction	Low to Med	Position as identity/placemaking first; DOOH only if owner wants it

16. Next-Sprint Execution Pack
(Outreach Plan, Prequal Checklist, Data Room Requests, Meeting Agenda)

Key Takeaway:
You need a highly disciplined "first contact package" that demonstrates you are mockup-ready and interface-ready, not just a product vendor.

A. Stakeholder Outreach plan (who to contact first)

- JANGHO (Façade Carrier)
- FMS (Acceptance Gate)
- AS+GG (Design Gate)
- TURNER (Process/Procurement Visibility)
- SBG (Commercial Award Gate)
- DAR (Local Compliance Reviewer)

B. Prequalification Checklist (SAVC + Consortium)

- Company Profile, Financials, Global Project References
- QA/QC: ISO, ITP templates, factory acceptance test (FAT) procedure
- Product documentation: datasheets, wiring diagrams, photometrics, IP ratings
- Reliability: MTBF assumptions, burn-in, spare parts policy
- Warranty + SLA plan
- Cyber/security posture for CMS/network
- HSE plan + method statements for façade works

C. Data Room Request List (Minimum)

- Façade drawings: typologies, mullion details, spandrel zones, access constraints
- Curtain wall sequencing and shop drawing timeline
- Lighting narrative/specs from FMS
- Architectural façade elevations and night intent renderings
- Submittal workflow + approval matrix (TURNER)
- Any prior façade lighting/media façade proposals/submittals [To verify]
- Constraints: glare limits, content rules, event mode expectations

D. Meeting Agenda (JANGHO Workshop)

- Panel typologies & permitted penetrations
- Cable routing & compartmentalization
- Service access strategy
- Mockup definition + acceptance criteria
- Delivery schedule alignment
- Roles/responsibilities + submittal routing

17. Go / No-Go Decision Gate (Based on Access, Timing, Budget, Exclusivity)

Key Takeaway:
You should Go only if you can secure (1) a real technical workshop with JANGHO, and (2) an acceptance path with FMS/AS+GG, within the next sprint; otherwise the probability of late-stage rejection rises sharply.

GO if all are achieved within 30 to 45 days

- Access:
  Confirmed workshops with JANGHO + FMS and a named TURNER/SBG procurement interface.

- Timing:
  Curtain wall integration window still open (shop drawing / fabrication freeze not passed) [To verify via JANGHO]

- Budget intent:
  Owner accepts façade lighting/media budget band (even if phased).

- Exclusivity leverage:
  JANGHO agrees to treat SAVC as nominated subsystem partner (non-exclusive is acceptable, but nomination must be real).

NO-GO if any two of the following are true

- No access to FMS/AS+GG acceptance pathway
- JANGHO declines integration or insists on their own nominated tech
- SBG procurement cannot confirm package buyability or blocks vendor onboarding

18. Contact Entry Points and Internal Ownership

ENTITY	ROLE IN DECISION	ENTRY POINT	OUTREACH OWNER	PRIORITY
JEC / Kingdom	Owner/developer; approves scope and brand intent	https://kingdom.com.sa/contact-us	Local / Vasee	P1
SBG	Principal contractor; procurement execution; site access	https://www.sbg.com.sa/contact-us/	Local / Vasee	P1
Jangho / Jangho Arabia	Facade subcontract; embedded subsystem gate	https://janghofacade.com/contact/ and https://en.jangho.com/JANGHONEWS/2025/1203/140.html	SAVC Leadership	P1
Turner	Program management; governance; routing	https://www.turner-arabia.com/contact/	SAVC Leadership	P1
DAR	Engineering support; compliance acceptance	https://www.dar.com/contact-us/	SAVC Technical Team	P1
AS + GG	Architect; design intent and BoD influence	https://smithgill.com/contact	SAVC Design Team	P2
FMS	Lighting consultant; performance criteria; mock-up acceptance	https://www.fms-lt.com/contact/	SAVC Design Team	P2

19. References

A. Primary / Authoritative (Project restart + named delivery chain)

1. Kingdom Holding / Jeddah Economic Company (JEC) – Restart agreement (JEC-SBG; SAR 7.2B; 42 months)
   https://kingdom.com.sa/news/jeddah-economic-company-has-signed-an-agreement

2. KONE (Press Release) – Confirms stakeholder spine: JEC (developer), SBG (principal contractor), Turner (PM), DAR (engineering support), AS+GG (architecture)
   https://www.kone.com/en/news-and-insights/releases/kone-to-equip-jeddah-tower--the-world-s-tallest-building-to-be-in-saudi-arabia-with-people-flow--solutions-2025-10-08.aspx

3. KONE (Regional mirror) – Same stakeholder confirmation
   https://www.kone.bh/en/news-insights/press-releases/kone-partners-with-jeddah-tower.aspx\

B. Project Database / Stakeholder Confirmation

1. CTBUH SkyscraperCenter – Jeddah Tower listing (Owner/Developer: JEC; Design Architect: AS+GG; AOR: DAR; Lighting: Fisher Marantz Stone; Wayfinding: Forcade)
   https://www.skyscrapercenter.com/building/jeddah-tower/2

C. Façade Package / Execution (Jangho disclosures)

1. Jangho Façade – “Rebuilding” / technical services narrative (Jangho Arabia mentioned; tests referenced)
   https://www.janghofacade.com/page91

2. Jangho Façade – Project page including curtain wall area claim (160,000 m^2) and unitized curtain wall description
   https://www.janghofacade.com/page597

D. Secondary News / Context (used for timeline triangulation only)

1. Gulf News – Restart coverage referencing 42-month timeline (secondary)
   https://gulfnews.com/business/property/jeddah-tower-saudi-firm-resumes-work-on-worlds-tallest-tower-1.1727944976026

E. Industry / Reference Context (used in partner/context sections)

1. Huda Lighting – Riyadh showroom (example of local support/distribution footprint reference)
   https://www.hudalighting.com/the-showroom-riyadh/index.html

F. Competitor Evidence

1. SACO / Citiled by SACO – Digital Media Facades positioning (competitor evidence link)
   https://saco.com/dmf/

G. Additional links (included for completeness; not all were used as “verified claim” evidence in the current write-up)

1. MEED – Chinese firm wins façade work (background)
   https://www.meed.com/chinese-firm-wins-facade-work-on-worlds-tallest-tower

2. Chicago Architecture Center – Encyclopedia entry (background overview)
   https://www.architecture.org/online-resources/architecture-encyclopedia/jeddah-tower

3. DETAIL – Kingdom Tower article (background)
   https://www.detail.de/de_en/kingdom-tower-in-jeddah-25652?srsltid=AfmBOoo5ofArpTjWNUuAoqO_3F7wN9sVNwfL3Hyp3qpBRGiOzjOokZcF

4. AS+GG – Jeddah Tower project page (background / design reference)
   https://smithgill.com/work/jeddah_tower/

5. FacadeLighting.com – Saudi Arabia page (background vendor/industry directory)
   https://www.facadelighting.com/saudi-arabia

6. Prisme Entertainment – references page (background)
   https://prismeent.com/references/